

DIGITAL IMAGE PROCESSING MINI PROJECT REPORT

Coin Detection and Counting Using Image Segmentation Techniques

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Abstract

This mini project presents a coin detection and counting system based on digital image processing and image segmentation techniques. The implemented pipeline uses grayscale conversion, median filtering, histogram analysis, thresholding, Otsu's thresholding, morphological processing, contour detection, and area-based filtering. Three test images were evaluated. The system detected 15/15 coins in the first image, 58/57 in the second, and 9/6 in the overlapping-coin image. Average counting agreement accuracy was 82.75%.

1. Introduction

Digital Image Processing provides techniques for improving and analysing images. Image segmentation separates meaningful objects from background. This project applies segmentation and contour analysis to the practical problem of automatic coin counting.

2. Problem Statement

Develop an image-processing application that detects and counts coins from digital images using segmentation and contour analysis, including preprocessing, binary segmentation, morphological processing, contour detection, filtering, and counting.

3. Objectives

Understand image segmentation; preprocess coin images; compare global and Otsu thresholding; improve segmentation using morphology; detect contours; filter small regions; count coins; evaluate results; identify limitations and improvements.

4. Dataset and Image Description

A small test dataset of three coin images was used: dip3.jpeg for a basic separated-coin test; international-coins.jpg for a normal multi-coin test; and famous-heads-coin-collection.jpg as a challenging overlapping-coin test. The images were collected from public web image sources.

5. Tools and Technologies

Python, OpenCV, NumPy, Matplotlib, Pandas, Google Colab, and GitHub were used.

6. Methodology

Pipeline: Input Image → Grayscale → Median Filtering → Histogram → Thresholding/Otsu → Morphological Processing → Contour Detection → Area Filtering → Counting → Evaluation. A 5×5 median filter and 5×5 morphology kernel were used. Otsu thresholding was the main automatic segmentation method.

7. Implementation

The implementation was developed incrementally in Google Colab using OpenCV functions including cv2.imread(), cv2.cvtColor(), cv2.medianBlur(), cv2.threshold(), cv2.morphologyEx(), cv2.findContours(), cv2.contourArea(), and cv2.boundingRect().

8. Results

The actual counts were manually determined and compared with detected counts from the contour-based system.

Image	Actual	Detected	Error	Accuracy (%)
dip3.jpeg	15	15	0	100.00

Image	Actual	Detected	Error	Accuracy (%)
international-coins.jpg	57	58	1	98.25
famous-heads-coin-collection.jpg	6	9	3	50.00

Average counting agreement accuracy: 82.75%

Total absolute counting error: 4 coins.

9. Evaluation and Interpretation

Counting agreement accuracy was calculated as $(1 - |\text{Actual} - \text{Detected}| / \text{Actual}) \times 100$. The metric describes count agreement and is not equivalent to precision, recall, or segmentation IoU. The first image achieved 100%, the second 98.25%, and the overlapping image 50%.

10. Comparison of Thresholding Approaches

Global thresholding uses a manually selected fixed value, while Otsu thresholding automatically selects a value from the intensity distribution. Otsu was used as the main thresholding approach.

11. Limitations

The method works best for separated coins. Touching/overlapping coins, illumination changes, shadows, reflections, similar foreground/background intensity, and parameter selection can reduce accuracy. Only three images were evaluated.

12. Future Scope

Watershed segmentation, adaptive thresholding, Hough Circle Transform, denomination classification, larger datasets, and real-time camera processing can improve the system.

13. Conclusion

The project demonstrates a practical coin counting application using grayscale conversion, filtering, thresholding, Otsu's method, morphology, contour detection, and area filtering. It performs well on separated coins and shows clear limitations for overlapping coins.

14. Project Files and GitHub Structure

source_code.py; requirements.txt; coin_counting_results.csv; images/; screenshots/; report/; README.md.

15. References

OpenCV documentation; NumPy documentation; Matplotlib documentation; Pandas documentation; course-provided Digital Image Processing Mini Project Guidelines; public web image sources used for the selected test images.